

Scalar first order LDE's

$$(1) \quad x' + a(t)x = f(t), \quad t \in I$$

where $a, f \in C(I)$.

For $t_0 \in I$ fixed, consider $A(t) = - \int_{t_0}^t a(s) ds$. Thus $A'(t) = -a(t)$
 $A(t_0) = 0$.

I. The integrating factor method to find the general sol of (1)

Theorem 1 The function $\mu: I \rightarrow \mathbb{R}$, $\mu(t) = e^{A(t)}$ is an integrating factor for (1).

The general sol. for (1) is

$$(2) \quad x = c \cdot e^{A(t)} + e^{A(t)} \cdot \int_{t_0}^t f(s) e^{-A(s)} ds, \quad c \in \mathbb{R} \text{ arbitrary.}$$

Proof

$$x' + a(t)x = f(t) \quad | \cdot e^{-A(t)} \Leftrightarrow \underbrace{x' e^{-A(t)} + x \cdot a(t) e^{-A(t)}}_{(x \cdot e^{-A(t)})'} = f(t) e^{-A(t)}$$

$$\Leftrightarrow (x \cdot e^{-A(t)})' = f(t) \cdot e^{-A(t)}, \quad \forall t \in I$$

integrate

$$x \cdot e^{-A(t)} = \int_{t_0}^t f(s) \cdot e^{-A(s)} ds + c, \quad \forall t \in I \quad \Rightarrow \quad x = c \cdot e^{A(t)} + e^{A(t)} \cdot \int_{t_0}^t f(s) e^{-A(s)} ds$$

$\forall t \in I$
where $c \in \mathbb{R}$ arbitrary.

II. The separation of variables method to find the general sol. of $x' + a(t)x = 0$

Def. A DE of the form $x' = \underbrace{b(t)} \cdot \underbrace{g(x)}$ is said to be a separable eq.

$$(x' \rightarrow \frac{dx}{dt} \quad \frac{dx}{dt} = b(t) \cdot g(x) \quad \frac{dx}{g(x)} = b(t) dt \quad \text{we separated the variables})$$

Notice that $x' + a(t)x = 0$ can be written as $x' = -a(t)x$, thus is a separable equation.

Remark. We already know that $x = c \cdot e^{A(t)}$, $c \in \mathbb{R}$ is the general sol.

From here we deduce that $x = 0$ is a (constant) solution and any other sol.

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From here we deduce that $x=0$ is a (constant) solution and any other sol. satisfies $x(t) \neq 0 \quad \forall t \in I$.

The sep. of var. method: $x'(t) = -a(t)x(t)$ $x=0$ is a sol.

Now we look for sol. $x \neq 0$ $\frac{x'(t)}{x(t)} = -a(t) \Leftrightarrow (\ln|x(t)|)' = -a(t)$

integrate $\Leftrightarrow \ln|x(t)| = A(t) + c, \quad c \in \mathbb{R} \Leftrightarrow |x(t)| = e^{A(t)+c} \Leftrightarrow$

$\Leftrightarrow x(t) = \pm e^c \cdot e^{A(t)}, \quad c \in \mathbb{R}$ arbitrary

Note that

Recall $x=0$ sol.

$\{e^c, -e^c, 0 : c \in \mathbb{R}\} = \mathbb{R}$

$\Leftrightarrow x(t) = c_1 \cdot e^{A(t)}, \quad c_1 \in \mathbb{R}$ arbitrary

Remark. we can use the sep. of var. method to find just a solution

$x_1 \neq 0$. How? when we integrate we take $c=0$, that is

$\ln|x(t)| = A(t) \Rightarrow x_1(t) = e^{A(t)}$ this is a sol. of a first order LODE

we apply the fundamental th. for LHDE and obtain that the general sol. is $x = c \cdot e^{A(t)}, \quad c \in \mathbb{R}$.

III The Lagrange method (the variation of constant method) to

find a particular solution of (1) $x' + a(t)x = f(t)$ (linear nonhom.)

The idea of Lagrange was to look at the general sol. of the LHDE

$x' + a(t)x = 0$, which is $x_h = c \cdot e^{A(t)}, \quad c \in \mathbb{R}$ arbitrary

and, then, to look for a partic. sol. of (1) of the form

$x_p = \varphi(t) \cdot e^{A(t)}, \quad \varphi = ?$

x_p is a sol. of (1) $\Leftrightarrow \varphi'(t) \cdot e^{A(t)} + \varphi(t) \cdot e^{A(t)} (-a(t)) + a(t) \cdot \varphi(t) \cdot e^{A(t)} = f(t)$

$\Leftrightarrow \varphi'(t) = f(t) \cdot e^{-A(t)} \Leftrightarrow \varphi(t) = \int_{t_0}^t f(s) e^{-A(s)} ds, \quad \forall t \in I$

so $|x| \leq e^{A(t)} \int_{t_0}^t f(s) e^{-A(s)} ds$

$$\text{So, } \left| x_p = e^{A(t)} \int_{t_0}^t f(s) e^{-A(s)} ds \right|$$

Remark. Thus, the general sol. of (1) is $x = x_h + x_p$, that is we found again (2).

Exercises Find the general sol. of

a) $x' - 2tx = t$, $t \in \mathbb{R}$

b) $x' - 2tx = \frac{2}{\sqrt{t}}$, $t \in \mathbb{R}$

(HW) c) $x' + \frac{1}{t}x = \frac{1}{t}e^{-2t+1}$, $t \in (0, \infty)$

a) $x' - 2tx = t$ first order L nonhom DE

Step 1 we write the LHD associated $x' - 2tx = 0$
 apply the sep. of var. method $x' = 2tx$ $\frac{dx}{dt} = 2tx$ $\int \frac{dx}{x} = \int 2t dt$
 $\ln|x| = t^2 + c$, $|x| = e^{t^2+c}$, $x = \pm e^c \cdot e^{t^2}$ $\Rightarrow x_h = c_1 \cdot e^{t^2}$, $c_1 \in \mathbb{R}$
 $x = 0$

Step 2 $x_p = ?$ Note that $x_p = -\frac{1}{2}$ is a sol.

Step 3 the gen. sol. $x = c_1 \cdot e^{t^2} - \frac{1}{2}$, $c_1 \in \mathbb{R}$

b) $x' - 2tx = \frac{2}{\sqrt{t}}$

Step 1 it has the same hom. part as a). Thus $x_h = c_1 \cdot e^{t^2}$, $c_1 \in \mathbb{R}$

Step 2. $x_p = ?$ we apply the Lagrange method

$$x_p = \varphi(t) \cdot e^{t^2}$$

$$x_p \text{ is a sol. } \Leftrightarrow \varphi'(t) \cdot e^{t^2} + \varphi(t) \cdot e^{t^2} \cdot 2t - 2t \varphi(t) e^{t^2} = \frac{2}{\sqrt{t}} \Leftrightarrow$$

$$\Leftrightarrow \varphi'(t) = \frac{2}{\sqrt{t}} e^{-t^2} \Leftrightarrow \varphi(t) = \frac{2}{\sqrt{\pi}} \int_0^t e^{-s^2} ds$$

it is known that
 the primitive of
 $-t^2$

The Gauss error function

The Gauss error function

$$\operatorname{erf}(t) \stackrel{\text{not}}{=} \frac{2}{\sqrt{\pi}} \int_0^t e^{-s^2} ds, \quad \forall t \in \mathbb{R}$$

So, $\varphi(t) = \operatorname{erf}(t) \Rightarrow x_p = \operatorname{erf}(t) \cdot e^{t^2}$

Step 3 $x = c \cdot e^{t^2} + \operatorname{erf}(t) \cdot e^{t^2}, \quad c \in \mathbb{R} \text{ arbitrary.}$

the primitive of e^{-t^2} is not a finite combination of elementary functions, thus we are not able to write its expression.

Find an integrating factor for a) and b). It should be the same since both eq. have the same homogeneous part.

$$\underbrace{x' - 2tx = t}_{\text{homogeneous part}}$$

$$a(t) = -2t$$

$$A(t) = -\int_0^t a(s) ds = 2 \int_0^t s ds = s^2 \Big|_0^t = t^2$$

$$\mu(t) = e^{-A(t)} = e^{-t^2} \quad \text{an integrating factor}$$

$$x' - 2tx = t \mid \cdot e^{-t^2} \Leftrightarrow x' \cdot e^{-t^2} - 2tx e^{-t^2} = t \cdot e^{-t^2} \Leftrightarrow$$

$$\Leftrightarrow (x \cdot e^{-t^2})' = t e^{-t^2} \quad \Leftrightarrow \quad x \cdot e^{-t^2} = -\frac{1}{2} e^{-t^2} + c, \quad c \in \mathbb{R} \quad \Leftrightarrow$$

$$\left(\int t e^{-t^2} dt = -\frac{1}{2} \int (e^{-t^2})' dt = -\frac{1}{2} e^{-t^2} \right)$$

$$\Leftrightarrow \boxed{x = c \cdot e^{t^2} - \frac{1}{2}, \quad c \in \mathbb{R}.}$$

Remark. on the Gauss error function

$$\operatorname{erf}(t) = \frac{2}{\sqrt{\pi}} \int_0^t e^{-s^2} ds.$$

Since $\int_0^\infty e^{-s^2} ds = \frac{\sqrt{\pi}}{2}$ we have $\lim_{t \rightarrow \infty} \operatorname{erf}(t) = 1$.

It is easy to prove that erf is an odd function, i.e. $\operatorname{erf}(-t) = -\operatorname{erf}(t), \quad \forall t \in \mathbb{R}$

Thus we have $\lim_{t \rightarrow -\infty} \operatorname{erf}(t) = -1$.

Also, note that $\operatorname{erf}(0) = 0$, erf is a strictly increasing function (since $\operatorname{erf}'(t) = \frac{2e^{-t^2}}{\sqrt{\pi}}$)

Thus its graph looks like

